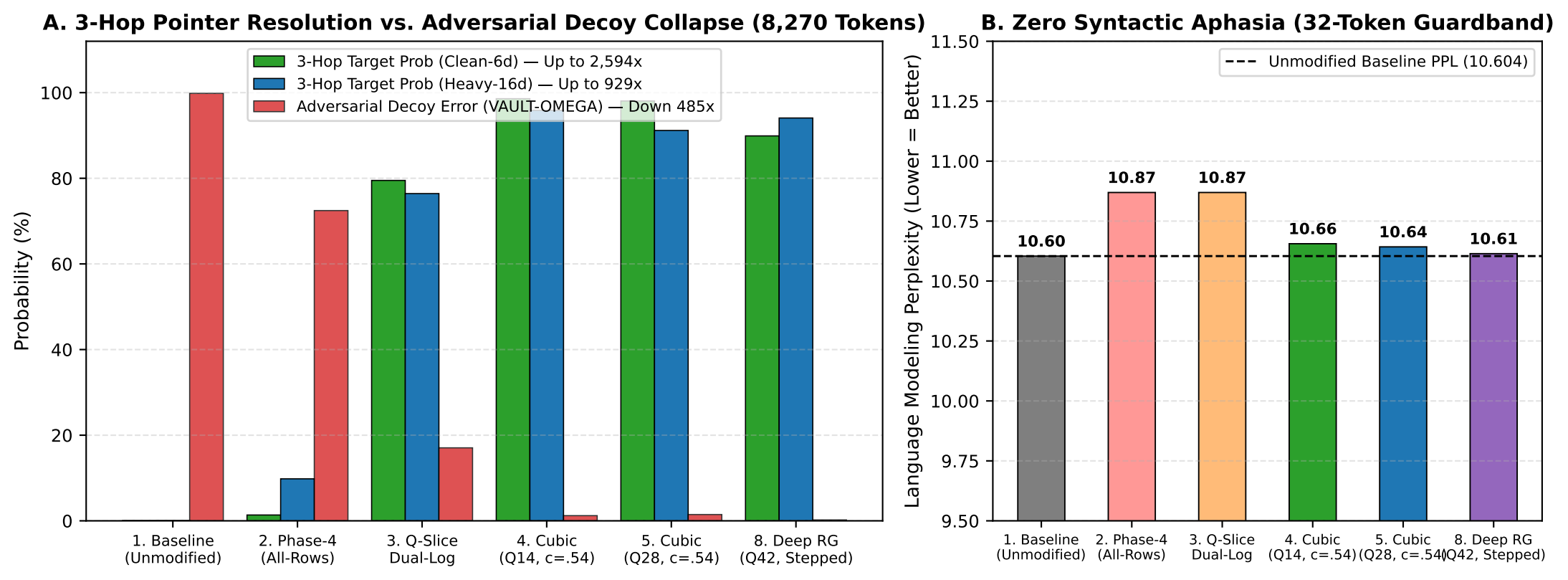
A. 3-Hop Pointer Resolution vs. Adversarial Decoy Collapse (8,270 Tokens)
3-Hop Target Prob (Clean-6d) — Up to 2,594x
3-Hop Target Prob (Heavy-16d) — Up to 929x
Adversarial Decoy Error (VAULT-OMEGA) — Down 485x
Probability (%)
0
20
40
60
80
100
1. Baseline (Unmodified)
2. Phase-4 (All-Rows)
3. Q-Slice Dual-Log
4. Cubic (Q14, c=.54)
5. Cubic (Q28, c=.54)
8. Deep RG (Q42, Stepped)
B. Zero Syntactic Aphasia (32-Token Guardband)
Unmodified Baseline PPL (10.604)
Language Modeling Perplexity (Lower = Better)
9.50
9.75
10.00
10.25
10.50
10.75
11.00
11.25
11.50
10.60
10.87
10.87
10.66
10.64
10.61
1. Baseline (Unmodified)
2. Phase-4 (All-Rows)
3. Q-Slice Dual-Log
4. Cubic (Q14, c=.54)
5. Cubic (Q28, c=.54)
8. Deep RG (Q42, Stepped)